

Visual Acuity and Cardiovascular Mortality in 2.52 Million Older Adults in China

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ABSTRACT

BACKGROUND Visual acuity (VA) impairment and cardiovascular disease (CVD) are major global health burdens. Although impaired VA is linked to elevated all-cause mortality risk, its association with CVD mortality, particularly in older population, remains inadequately explored and inconsistent.

OBJECTIVES This study aimed to quantify the burden of VA impairment and investigate its association with CVD mortality in a large cohort of Chinese older adults.

METHODS This multicenter cohort study used data from the Basic Public Health Services in China. A total of 2,517,662 participants 65 years of age or older from 4 cities (Zaozhuang, Luzhou, Zunyi, and Shenzhen) were included between 2017 and 2023. Presenting visual acuity of the better-seeing eye (BVA) was measured using a standard logarithmic tumbling E chart. Mortality data were ascertained via linkage to the China Population Death Information Registration System, with follow-up through December 31, 2024. Cox proportional hazards models were used to estimate HRs and 95% CIs for the association between VA and mortality, with adjustments for demographic, lifestyle, clinical and geographic factors. Stratified analysis, competing risk analysis, dose-response analysis, and extensive sensitivity analyses were performed to further investigate the relationship.

RESULTS Among the 2.52 million participants (mean age: 70.67 years), 42.5% had VA impairment. Over a mean follow-up of 4.99 years (accumulating 12,560,718.9 person-years), 348,501 deaths occurred, including 153,758 CVD deaths. In the fully adjusted model, each 1-logMAR unit increase in BVA (greater visual impairment) was associated with a 3% higher risk of all-cause mortality (HR: 1.03; 95% CI: 1.03-1.04) and a 3% higher risk of CVD mortality (HR: 1.03; 95% CI: 1.02-1.03). A graded association was observed when VA was categorized by quartiles, with the worst quartile (Q4) showing significantly elevated risks for all-cause (HR: 1.25; 95% CI: 1.24-1.26) and CVD mortality (HR: 1.30; 95% CI: 1.28-1.32) compared to the best quartile (Q1). Stratified analysis indicated that the association was stronger in younger participants (65-70 years). Dose-response analysis revealed a significant nonlinear association, with mortality risk rising steeply at lower levels of impairment and plateauing at higher levels. Competing risk and sensitivity analysis confirmed the robustness of these findings.

CONCLUSIONS Beyond its traditional ocular role, VA impairment emerges as a sentinel marker of cardiovascular vulnerability and a functional barometer of biological aging. Situating cardiovascular risk within the broader landscape of functional aging and prioritizing holistic functional preservation could be pivotal for promoting longevity and achieving healthy aging goals. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****BVA** = the presenting visual
acuity of the better-seeing eye**CVD** = cardiovascular disease**ICD-10** = International
Classification of Diseases, 10th
Revision**MVA** = mean binocular
presenting visual acuity**sHR** = subdistribution HR**VA** = visual acuity**WVA** = the presenting visual
acuity of the worse-seeing eye

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide.¹ Although extensive research has established the pivotal role of traditional risk factors such as hypertension, diabetes, dyslipidemia, and smoking, there is growing recognition that a substantial portion of CVD risk stems from nontraditional factors.² In this context, visual acuity (VA) represents a compelling yet underexplored candidate for systemic cardiovascular health monitoring. Affecting >2.2 billion people globally,³ VA impairment carries systemic implications that extend far beyond

localized ocular dysfunction.⁴ The eye's intricate microvascular and neural architecture provides a unique, noninvasive window into the body's internal environment, reflecting the cumulative burden of vascular injury and biological aging.⁵⁻⁷ Particularly, VA impairment shares common pathological pathways with CVD, including chronic inflammation, endothelial dysfunction, and traditional metabolic stressors.⁸⁻¹¹ Furthermore, VA impairment could precipitate a cascade of secondary stressors, including physical inactivity, social isolation, psychological distress, and cognitive decline.⁴ Thus, compared with traditional cardiovascular risk factors, VA could operate as a more integrative holistic marker: it not only captures the life-course accumulation of vascular damage and biological aging but also reflects an individual's functional reserve, social context, and access to health care.^{4,7}

Although impaired VA has been linked to elevated all-cause mortality,¹² the evidence concerning its specific association with CVD mortality remained limited and inconsistent.¹³⁻¹⁶ For instance, the Singapore Malay Eye Study identified impaired VA as a significant marker for increased CVD mortality.¹³ However, recent analyses of the U.S. National Health and Nutrition Examination Survey and the Central Australian Ocular Health Study found no statistically significant link between VA impairment and CVD mortality.^{14,16} Conversely, the Kangbuk Samsung Health Study showed a clear dose-response relationship between VA impairment and CVD mortality risk among a young and middle-aged Korean population,¹⁵ further highlighting the inconsistency in the existing evidence. Against the backdrop of a rapidly aging Chinese population and the anticipated concurrent rise in VA impairment and CVD burden among older adults, the relationship between VA and CVD mortality within this specific demographic remains unexamined. Moreover, even basic large-scale epidemiological data detailing the burden of VA

impairment in China are notably scarce. Additionally, no prior study has specifically investigated whether the association between VA and mortality varies by different VA metrics, such as the VA of the better-seeing eye vs the worse-seeing eye.

To address these critical knowledge gaps, this study aims to achieve the following objectives: first, to quantify the burden of VA impairment among Chinese older adults; second, to systematically evaluate its association with CVD mortality using multicenter, city-level, large-scale data; and third, to investigate whether this association differs according to specific VA metrics, such as the VA of the better-seeing eye vs the worse-seeing eye. The findings from this investigation are anticipated to inform the development of integrated public health strategies aimed at preserving VA and preventing CVD within aging populations.

METHODS

STUDY POPULATION. The current study used data from the Basic Public Health Services in China, a government-initiated project that provides free, fundamental public health services.^{17,18} Particularly, the Health Management Program for the Elderly is a core component of the project. The program provides an annual, complimentary health management service to all residents 65 years of age and older, encompassing both urban and rural populations. This comprehensive service package includes: 1) a thorough assessment of lifestyle and health status; 2) a detailed physical examination; 3) a suite of auxiliary investigations; and 4) tailored health guidance. This structured initiative is designed to facilitate primary prevention and enable early detection of health abnormalities among the elderly.

The study population consisted of examinees who participated in the Basic Public Health Services project in 4 cities (Zaozhuang, Luzhou, Zunyi, and Shenzhen) between 2017 and 2023 (N = 2,935,867). There were 64,519 subjects who were excluded for missing VA data and 353,686 subjects who were excluded for missing covariates data. Ultimately, the final analysis dataset consists of 2,517,662 eligible participants. The detailed screening procedure of study participants is shown in [Supplemental Figure 1](#), whereas the baseline profiles for both included and excluded individuals are summarized in [Supplemental Table 1](#).

This study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of National Center for Chronic and Noncommunicable Disease Control and Prevention,

TABLE 1 Visual Acuity at a Glance^a

Category ^b	Snellen Fraction	logMAR Value	5-Point Value in China	Reading Performance ^c
Normal range	≥6/12	≤0.30	≥4.70	Normal reading speed and distance Has reserve capacity for small print
Mild impairment	≥6/18 to <6/12	>0.30 to ≤0.48	≥4.52 to <4.70	Normal speed but reduced reading distance No reserve for small print
Moderate impairment	≥6/60 to <6/18	>0.48 to ≤1.00	≥4.00 to <4.52	Near-normal only with appropriate aids Requires low power magnifiers or large print books
Severe impairment	<6/60	>1.00	<4.00	Slower than normal even with aids Requires high power magnifiers (restricted field) May rely on spot reading or transition entirely to non-visual sources

While a point value of 6/6 is taken to be the optimal normal vision, this study utilizes a category of ≥6/12 as the reference group ("normal range"). Additionally, the "severe impairment" group denotes visual acuity <6/60, encompassing individuals with blindness (defined as <3/60 per WHO criteria). Although these reading performance descriptions parallel the functional visual acuity classifications (normal range, mild impairment, moderate impairment, severe impairment), they do not strictly map to rigid numerical cutoffs for visual acuity. ^aPresenting visual acuity is defined as the visual acuity measured upon arrival, with or without habitual corrective lenses. ^bCategories were adapted from WHO criteria presented in the *World Report on Vision*.³ ^cReading performance categorization was adapted from the classification of functional vision by Colenbrander.²⁰

Chinese Center for Disease Control and Prevention (No. 202328-01).

VISUAL ACUITY. VA was measured in its presenting state upon arrival for examination (presenting VA), representing participants' vision with or without their habitual corrective lenses.^{3,15} The Chinese standard logarithmic tumbling E chart with 5-point scale values¹⁹ was used for VA assessment. The tumbling E chart effectively overcomes educational and language barriers by eliminating the need for Latin alphabet recognition. Its testing method remains highly accessible, requiring only the identification of the opening direction of E letters, which also reduces cognitive load and testing anxiety while ensuring accurate VA assessment. This simplicity makes the chart both practical and culturally appropriate for China's aging population. During examination, the chart was positioned at a standard distance of 5 m from the subject's eye or, alternatively, 2.6 m away with a vertical mirror to achieve the total required 5-m optical path through reflection. VA was recorded as a 5-point (*F*) value, which is extensively used in Chinese clinical practice. The 5-point (*F*) values were converted into standard equivalent logarithm of the Minimum Angle of Resolution (logMAR) units (*L*) in the current analysis with the algorithm: $L = 5 - F$.

As the universally accepted standard in international clinical research, the logMAR scale offers significant advantages over Snellen fractions due to its linear and continuous numerical properties. This facilitates the calculation of arithmetic means, allows for precise clinical stratification, and enables direct inclusion in multivariable statistical models as either a continuous or categorical predictor. For

interpretability, it should be noted that higher logMAR values indicate poorer VA; for instance, a logMAR of 0 corresponds to a Snellen fraction of 6/6, whereas 1.0 corresponds to 6/60. The detailed mapping between the Chinese 5-point scale, logMAR units, and Snellen equivalents is provided in [Table 1](#).²⁰

OUTCOME. Participants' vital status was ascertained through linkage to the China Population Death Information Registration System using unique resident identification numbers,²¹ with follow-up censored at December 31, 2024. This system covers 31 provincial-level administrative regions in China and provides information on the date of death and cause of death for deceased population. All causes of death were coded according to the International Classification of Diseases, Tenth Revision (ICD-10). In the current study, CVD mortality was defined as underlying cause of death from diseases of the circulatory system, corresponding to ICD-10 codes I00-I99. Cancer mortality was identified using ICD-10 codes C00-C97 and D00-D09.

COVARIATES. Demographic characteristics, lifestyle factors, and medical history were collected with a standardized questionnaire. Blood pressure, height, weight and waist circumference were measured during physical examination. Laboratory tests included total cholesterol, triglycerides, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, fasting blood glucose, and glycated hemoglobin. Body mass index (BMI) was calculated and classified into 4 categories: underweight (<18.5 kg/m²), normal weight (≥18.5 to <25 kg/m²), overweight (≥24 to <28 kg/m²), and obesity (≥28 kg/m²) according to Chinese criteria.²² Hypertension was defined as systolic blood pressure ≥140 mm Hg

TABLE 2 Baseline Characteristics of the Study Participants by BVA Categories Based on WHO Visual Impairment Criteria

	Overall (N = 2,517,662)	Normal Range (≤ 0.3) (n = 1,446,625)	Mild Impairment (>0.3 to ≤ 0.48) (n = 343,926)	Moderate Impairment (>0.48 to ≤ 1.0) (n = 634,338)	Severe Impairment (>1.0) (n = 92,773)	P Value
BVA	0.46 $\pm$ 0.75	0.16 $\pm$ 0.12	0.40 $\pm$ 0.00	0.66 $\pm$ 0.17	3.94 $\pm$ 1.08	<0.001
WVA	0.56 $\pm$ 0.85	0.26 $\pm$ 0.37	0.50 $\pm$ 0.40	0.78 $\pm$ 0.57	4.01 $\pm$ 1.06	<0.001
MVA	0.51 $\pm$ 0.78	0.21 $\pm$ 0.21	0.45 $\pm$ 0.20	0.72 $\pm$ 0.32	3.98 $\pm$ 1.06	<0.001
Year	4.99 $\pm$ 2.07	4.85 $\pm$ 2.00	5.03 $\pm$ 2.12	5.10 $\pm$ 2.15	6.16 $\pm$ 1.95	<0.001
Month	59.84 $\pm$ 24.80	58.22 $\pm$ 23.97	60.37 $\pm$ 25.38	61.21 $\pm$ 25.80	73.87 $\pm$ 23.43	<0.001
Person-years	12,560,718.9	7,021,681.6	1,730,792.2	3,236,943.6	571,301.5	
Mortality						
All-cause	348,501 (13.84)	142,758 (9.87)	53,514 (15.56)	131,199 (20.68)	21,030 (22.67)	
CVD	153,758 (6.11)	60,256 (4.17)	23,552 (6.85)	59,850 (9.44)	10,100 (10.89)	
Cancer	68,951 (2.74)	35,585 (2.46)	10,248 (2.98)	20,044 (3.16)	3,074 (3.31)	
Age, y	70.67 $\pm$ 6.03	69.52 $\pm$ 5.20	71.19 $\pm$ 6.22	72.74 $\pm$ 6.92	72.66 $\pm$ 6.33	<0.001
Age group, y						<0.001
65-70	1,464,481 (58.17)	960,237 (66.38)	185,031 (53.80)	280,094 (44.16)	39,119 (42.17)	
70-75	510,067 (20.26)	271,864 (18.79)	74,105 (21.55)	139,971 (22.07)	24,127 (26.01)	
75-80	302,597 (12.02)	133,125 (9.20)	46,967 (13.66)	106,281 (16.75)	16,224 (17.49)	
≥ 80	240,517 (9.55)	81,399 (5.63)	37,823 (11.00)	107,992 (17.02)	13,303 (14.34)	
Sex						<0.001
Male	1,194,188 (47.43)	718,740 (49.68)	155,327 (45.16)	276,769 (43.63)	43,352 (46.73)	
Female	1,323,474 (52.57)	727,885 (50.32)	188,599 (54.84)	357,569 (56.37)	49,421 (53.27)	
Education level						<0.001
Low literate	710,522 (28.22)	335,859 (23.22)	110,214 (32.05)	234,075 (36.90)	30,374 (32.74)	
Primary education	1,107,015 (43.97)	616,126 (42.59)	158,458 (46.07)	288,876 (45.54)	43,555 (46.95)	
Secondary education	628,921 (24.98)	437,926 (30.27)	69,691 (20.26)	103,896 (16.38)	17,408 (18.76)	
Higher education	71,204 (2.83)	56,714 (3.92)	5,563 (1.62)	7,491 (1.18)	1,436 (1.55)	
Marital status						<0.001
Married	2,031,389 (80.69)	1,226,871 (84.81)	268,232 (77.99)	465,292 (73.35)	70,994 (76.52)	
Never married	41,867 (1.66)	20,807 (1.44)	6,623 (1.93)	12,698 (2.00)	1,739 (1.87)	
Widowed/divorced	444,406 (17.65)	198,947 (13.75)	69,071 (20.08)	156,348 (24.65)	20,040 (21.60)	
Smoking status						<0.001
Never	1822,677 (72.40)	1,038,037 (71.76)	251,742 (73.20)	464,704 (73.26)	68,194 (73.51)	
Current	567,877 (22.56)	322,998 (22.33)	78,414 (22.80)	144,814 (22.83)	21,651 (23.34)	
Quit	127,108 (5.05)	85,590 (5.92)	13,770 (4.00)	24,820 (3.91)	2,928 (3.16)	
Alcohol consumption						<0.001
Never	1,948,877 (77.41)	1,109,127 (76.67)	271,260 (78.87)	495,843 (78.17)	72,647 (78.31)	
Current	434,119 (17.24)	271,135 (18.74)	55,032 (16.00)	94,976 (14.97)	12,976 (13.99)	
Quit	134,666 (5.35)	66,363 (4.59)	17,634 (5.13)	43,519 (6.86)	7,150 (7.71)	
BMI, kg/m ²	23.26 $\pm$ 3.28	23.54 $\pm$ 3.25	23.10 $\pm$ 3.26	22.79 $\pm$ 3.30	22.59 $\pm$ 3.21	<0.001
BMI						<0.001
Underweight	131,343 (5.22)	60,559 (4.19)	19,266 (5.60)	45,353 (7.15)	6,165 (6.65)	
Normal weight	1,456,269 (57.84)	797,630 (55.14)	205,813 (59.84)	391,413 (61.70)	61,413 (66.20)	
Overweight	726,556 (28.86)	458,774 (31.71)	92,610 (26.93)	155,051 (24.44)	20,121 (21.69)	
Obesity	203,494 (8.08)	129,662 (8.96)	26,237 (7.63)	42,521 (6.70)	5,074 (5.47)	
Exercise frequency						<0.001
Daily	622,616 (24.73)	454,091 (31.39)	62,193 (18.08)	94,004 (14.82)	12,328 (13.29)	
Weekly	110,763 (4.40)	71,249 (4.93)	13,515 (3.93)	21,845 (3.44)	4,154 (4.48)	
Occasionally	75,843 (3.01)	50,514 (3.49)	8,468 (2.46)	14,621 (2.30)	2,240 (2.41)	
Never	1,708,440 (67.86)	870,771 (60.19)	259,750 (75.52)	503,868 (79.43)	74,051 (79.82)	
Hypertension						<0.001
No	1,127,922 (44.80)	661,576 (45.73)	154,190 (44.83)	273,073 (43.05)	39,083 (42.13)	
Yes	1,389,740 (55.20)	785,049 (54.27)	189,736 (55.17)	361,265 (56.95)	53,690 (57.87)	

Continued on the next page

TABLE 2 Continued

	Overall (N = 2,517,662)	Normal Range (≤ 0.3) (n = 1,446,625)	Mild Impairment (> 0.3 to ≤ 0.48) (n = 343,926)	Moderate Impairment (> 0.48 to ≤ 1.0) (n = 634,338)	Severe Impairment (> 1.0) (n = 92,773)	P Value
Diabetes						<0.001
No	2,077,359 (82.51)	1,184,617 (81.89)	285,705 (83.07)	528,744 (83.35)	78,293 (84.39)	
Yes	440,303 (17.49)	262,008 (18.11)	58,221 (16.93)	105,594 (16.65)	14,480 (15.61)	
Dyslipidemia						<0.001
No	1,705,929 (67.76)	958,120 (66.23)	236,338 (68.72)	443,105 (69.85)	68,366 (73.69)	
Yes	811,733 (32.24)	488,505 (33.77)	107,588 (31.28)	191,233 (30.15)	24,407 (26.31)	
CVD						<0.001
No	2,310,058 (91.75)	1,311,992 (90.69)	318,856 (92.71)	591,763 (93.29)	87,447 (94.26)	
Yes	207,604 (8.25)	134,633 (9.31)	25,070 (7.29)	42,575 (6.71)	5,326 (5.74)	
NSD						<0.001
No	2,508,366 (99.63)	1,441,148 (99.62)	342,787 (99.67)	632,079 (99.64)	92,352 (99.55)	
Yes	9,296 (0.37)	5,477 (0.38)	1,139 (0.33)	2,259 (0.36)	421 (0.45)	

Values are mean $\pm$ SD or n (%).

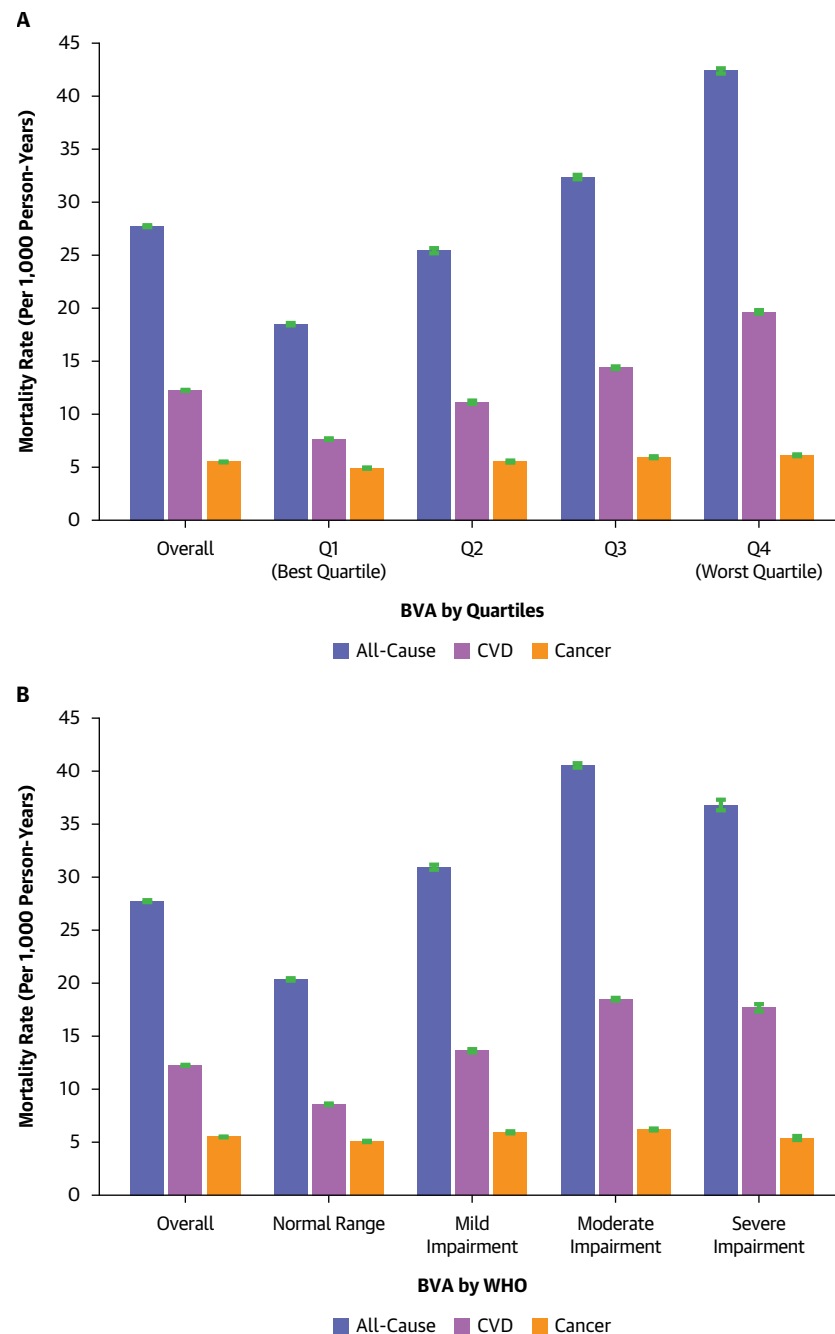
BMI = body mass index; BVA = the presenting visual acuity of the better-seeing eye; CVD = cardiovascular disease; MVA = mean binocular presenting visual acuity; NSD = nervous system disease; WVA = the presenting visual acuity of the worse-seeing eye.

and/or diastolic blood pressure ≥ 90 mm Hg, or a self-reported hypertension diagnosis.²³ Diabetes was defined as glycated hemoglobin $\geq 6.5\%$ and/or fasting blood glucose ≥ 7.0 mmol/L, or a self-reported diabetes diagnosis.²⁴ Dyslipidemia was defined as the presence of one or more of the following: total cholesterol ≥ 6.2 mmol/L, triglycerides ≥ 2.3 mmol/L, low-density lipoprotein cholesterol ≥ 4.1 mmol/L, high-density lipoprotein cholesterol ≤ 1.0 mmol/L, or a self-reported dyslipidemia diagnosis.²⁵ Covariates included age (65-70, 70-75, 75-80, ≥ 80 years), sex (male, female), education level (low-literate, primary education [≤ 6 years], secondary education [> 6 to ≤ 12 years], higher education [> 12 years]), marital status (married, never married, widowed/divorced), smoking status (never, quit, current), alcohol consumption (never, quit, current), BMI (underweight, normal weight, overweight, obesity), exercise frequency (daily, weekly, occasionally, never), hypertension (no, yes), diabetes (no, yes), dyslipidemia (no, yes), CVD (no, yes), and nervous system disease (no, yes) were ascertained according to baseline records. The ocular disease was defined as a self-reported diagnosis of eye conditions, ascertained via the baseline questionnaire with the question: "Do you have an ocular disease, or has a doctor ever told you that you have an ocular disease?"

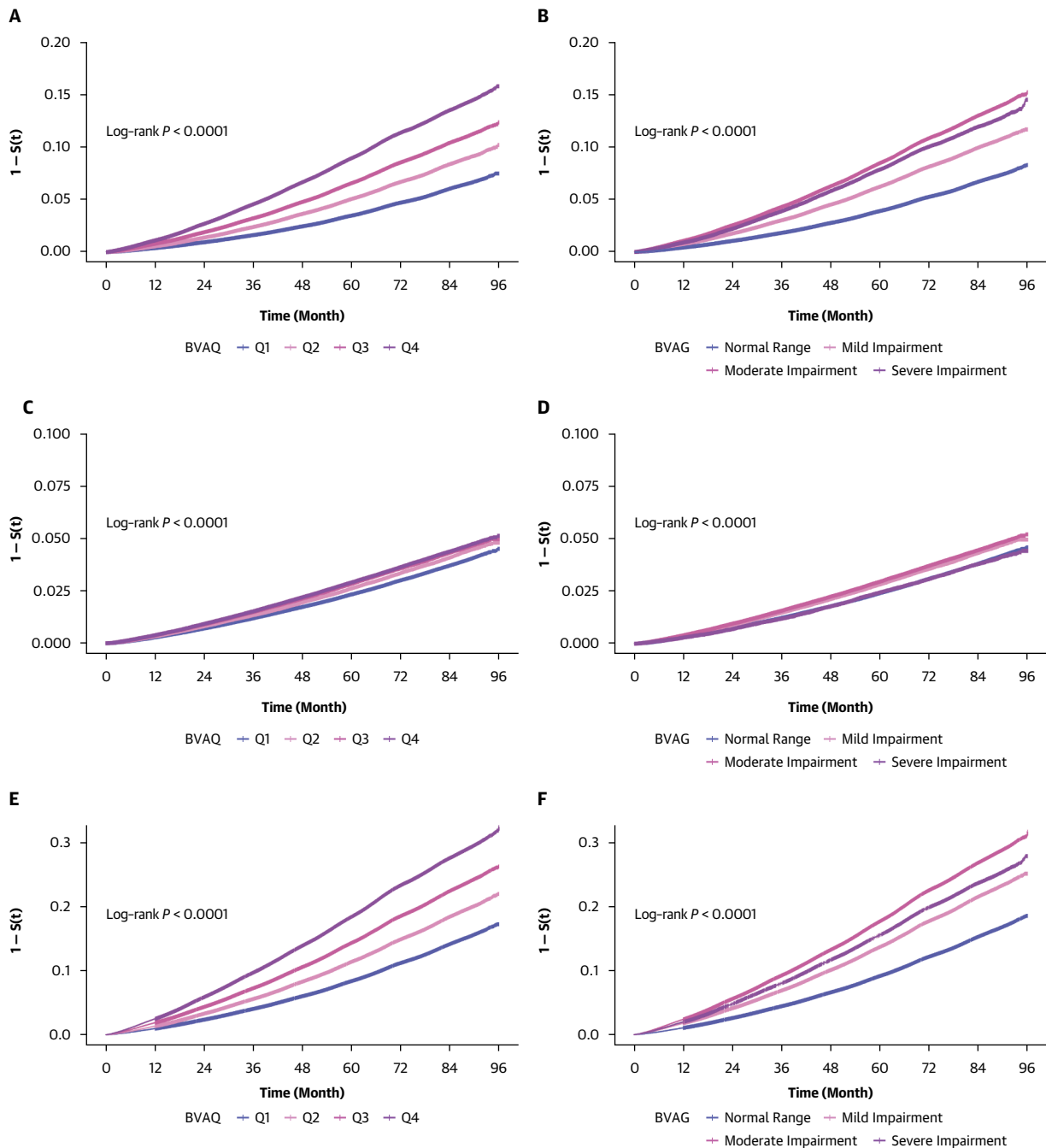
STATISTICAL ANALYSIS. The primary exposure was the presenting visual acuity of the better-seeing eye (BVA). In the current study, VA was analyzed as a continuous variable and in 2 categorical forms. The categorical forms were converted by 2 methods: 1) quartile-based grouping (Q1: ≤ 25 th percentile; Q2: > 25 th to ≤ 50 th percentile; Q3: > 50 th to ≤ 75 th

percentile; Q4: > 75 th percentile); and 2) categorization based on World Health Organization (WHO) VA impairment criteria (normal range: ≤ 0.3 ; mild impairment: > 0.3 to ≤ 0.48 ; moderate impairment: > 0.48 to ≤ 1.0 ; severe impairment: > 1.0).³ Functional descriptions for WHO categories are detailed in [Table 1](#), whereas specific logMAR quartile thresholds across different datasets are provided in [Supplemental Table 2](#). The primary outcomes included CVD mortality, cancer mortality, and all-cause mortality.

Baseline characteristics of the study population were presented according to the BVA groups. Continuous data were presented as mean $\pm$ SD, and categorical data as number (percentage). The association between VA and mortality were assessed using Cox proportional hazards models, with adjustments made in sequential stages: model 1 was unadjusted; model 2 was adjusted for age and sex; model 3 was further adjusted for socioeconomic and lifestyle factors (education level, marital status, smoking status, alcohol consumption, BMI, and exercise frequency); and model 4 was additionally adjusted for baseline clinical conditions (hypertension, diabetes, dyslipidemia, CVD, and nervous system disease) and geographic location (city of residence). To further investigate the relationship, we performed several additional analyses: 1) stratified analysis across subgroups (different age, sex, hypertension, diabetes, dyslipidemia, CVD, and city subgroups); 2) competing risk analysis using Fine-Gray proportional subdistribution hazards model for CVD mortality, treating deaths from other causes as competing events; 3) dose-response analysis using restricted cubic splines

FIGURE 1 Mortality Rates Across Different BVA Groups

(A) The presenting visual acuity of the better-seeing eye (BVA) categorized by quartiles. (B) BVA categorized based on WHO criteria. Crude mortality rates for cardiovascular disease (CVD), cancer, and all-cause mortality across different levels of visual acuity (VA) are compared. Higher quartiles (from Q1 to Q4) represent worsening visual acuity. Mortality rates for all-cause and CVD deaths generally exhibited a stepwise increase with worsening VA under quartile-based grouping. In contrast, cancer mortality rates remained relatively stable. These patterns suggest a specific link between visual acuity and cardiovascular health that differs from cancer outcomes. Q1 is ≤ 25 th percentile; Q2 is >25 th to ≤ 50 th percentile; Q3 is >50 th to ≤ 75 th percentile; and Q4 is >75 th percentile.

FIGURE 2 Kaplan-Meier Cumulative Mortality Probability Curves

Kaplan-Meier cumulative mortality probability curves [$1 - S(t)$] across different BVA groups: (A) BVA by quartiles and CVD mortality; (B) BVA by WHO and CVD mortality; (C) BVA by quartiles and cancer mortality; (D) BVA by WHO and cancer mortality; (E) BVA by quartiles and all-cause mortality; (F) BVA by WHO and all-cause mortality. Showing the cumulative probability of death over the follow-up period stratified by visual acuity categories. Higher quartiles (from Q1 to Q4) represent worsening VA. The curves show distinct risk stratification for CVD and all-cause mortality, with significantly higher mortality probability in participants with poorer VA under quartile-based grouping. In contrast, the curves for cancer mortality exhibit substantial overlap. The divergence in mortality curves specifically for CVD but not cancer supports the hypothesis that VA serves as a specific sentinel marker for cardiovascular vulnerability rather than cancer outcomes. BVAQ = BVA categorized by quartiles; BVAG = BVA categorized based on WHO criteria; other abbreviations as in [Figure 1](#).

TABLE 3 Associations of BVA With CVD Mortality, Cancer Mortality, and All-Cause Mortality

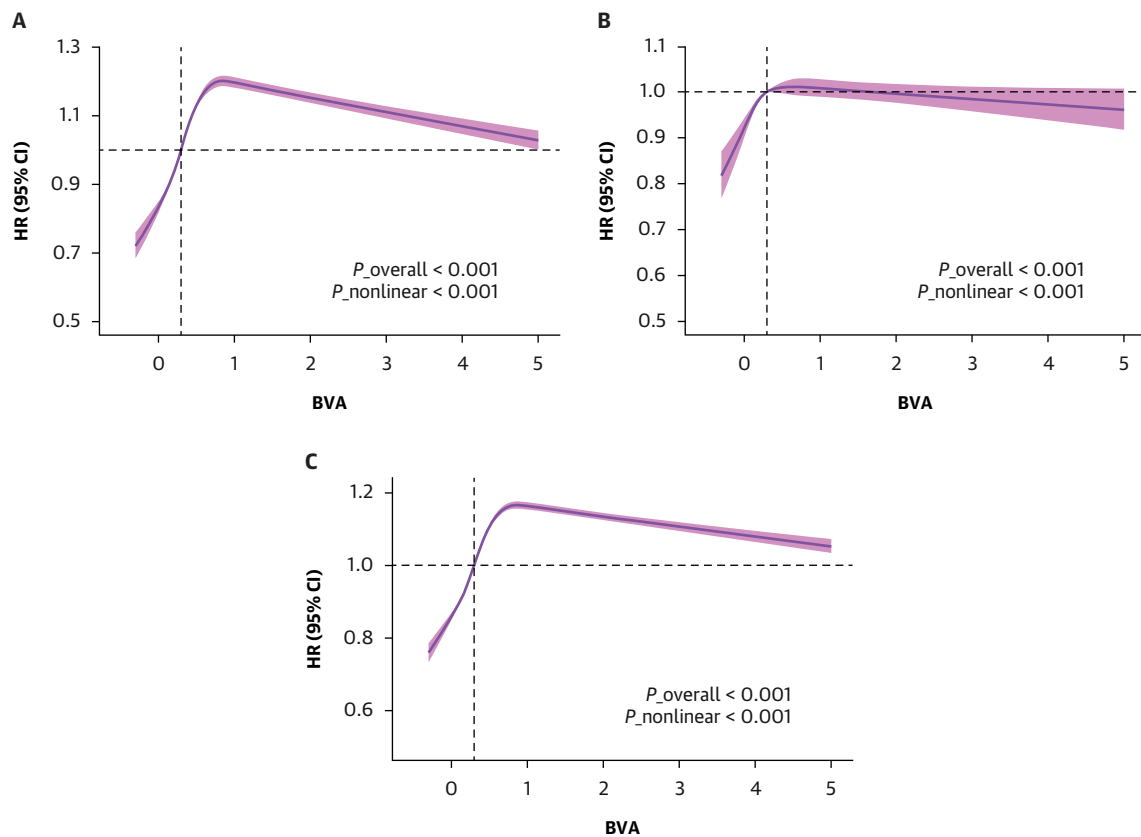
Exposure	Model 1 ^a		Model 2 ^b		Model 3 ^c		Model 4 ^d	
	HR (95% CI)	P Value	HR (95% CI)	P Value	HR (95% CI)	P Value	HR (95% CI)	P Value
CVD mortality								
BVA as continuous	1.16 (1.16-1.17)	<0.001	1.08 (1.07-1.08)	<0.001	1.05 (1.05-1.06)	<0.001	1.03 (1.02-1.03)	<0.001
BVA by quartiles								
Q1	Ref		Ref		Ref		Ref	
Q2	1.43 (1.41-1.46)	<0.001	1.22 (1.20-1.24)	<0.001	1.15 (1.13-1.17)	<0.001	1.13 (1.11-1.15)	<0.001
Q3	1.83 (1.81-1.86)	<0.001	1.39 (1.37-1.41)	<0.001	1.24 (1.22-1.26)	<0.001	1.20 (1.19-1.22)	<0.001
Q4	2.45 (2.42-2.49)	<0.001	1.57 (1.55-1.59)	<0.001	1.37 (1.35-1.39)	<0.001	1.30 (1.28-1.32)	<0.001
P for trend	<0.001		<0.001		<0.001		<0.001	
BVA by WHO								
Normal range	Ref		Ref		Ref		Ref	
Mild impairment	1.56 (1.53-1.58)	<0.001	1.26 (1.24-1.28)	<0.001	1.16 (1.14-1.18)	<0.001	1.13 (1.12-1.15)	<0.001
Moderate impairment	2.10 (2.07-2.12)	<0.001	1.44 (1.42-1.46)	<0.001	1.29 (1.27-1.30)	<0.001	1.24 (1.23-1.26)	<0.001
Severe impairment	1.89 (1.85-1.93)	<0.001	1.40 (1.37-1.43)	<0.001	1.27 (1.25-1.30)	<0.001	1.15 (1.13-1.18)	<0.001
P for trend	<0.001		<0.001		<0.001		<0.001	
Cancer mortality								
BVA as continuous	1.01 (1.00-1.02)	0.005	0.98 (0.97-0.99)	<0.001	0.97 (0.96-0.98)	<0.001	1.00 (0.99-1.01)	0.528
BVA by quartiles								
Q1	Ref		Ref		Ref		Ref	
Q2	1.11 (1.09-1.14)	<0.001	1.09 (1.06-1.11)	<0.001	1.06 (1.04-1.09)	<0.001	1.05 (1.03-1.08)	<0.001
Q3	1.19 (1.17-1.21)	<0.001	1.12 (1.10-1.14)	<0.001	1.07 (1.05-1.09)	<0.001	1.07 (1.05-1.09)	<0.001
Q4	1.21 (1.19-1.23)	<0.001	1.07 (1.05-1.10)	<0.001	1.02 (0.99-1.04)	0.139	1.05 (1.02-1.07)	<0.001
P for trend	<0.001		<0.001		0.004		<0.001	
BVA by WHO								
Normal range	Ref		Ref		Ref		Ref	
Mild impairment	1.15 (1.13-1.18)	<0.001	1.10 (1.08-1.12)	<0.001	1.07 (1.04-1.09)	<0.001	1.07 (1.04-1.09)	<0.001
Moderate impairment	1.20 (1.18-1.22)	<0.001	1.09 (1.07-1.10)	<0.001	1.04 (1.02-1.05)	<0.001	1.04 (1.02-1.06)	<0.001
Severe impairment	1.00 (0.96-1.03)	0.840	0.91 (0.87-0.94)	<0.001	0.87 (0.84-0.90)	<0.001	1.00 (0.96-1.04)	0.866
P for trend	<0.001		<0.001		0.417		0.002	
All-cause mortality								
BVA as continuous	1.14 (1.13-1.14)	<0.001	1.06 (1.06-1.06)	<0.001	1.04 (1.03-1.04)	<0.001	1.03 (1.03-1.04)	<0.001
BVA by quartiles								
Q1	Ref		Ref		Ref		Ref	
Q2	1.35 (1.34-1.37)	<0.001	1.19 (1.18-1.20)	<0.001	1.12 (1.11-1.13)	<0.001	1.10 (1.09-1.11)	<0.001
Q3	1.71 (1.69-1.72)	<0.001	1.35 (1.34-1.37)	<0.001	1.22 (1.21-1.23)	<0.001	1.17 (1.16-1.19)	<0.001
Q4	2.19 (2.17-2.21)	<0.001	1.50 (1.49-1.51)	<0.001	1.32 (1.30-1.33)	<0.001	1.25 (1.24-1.26)	<0.001
P for trend	<0.001		<0.001		<0.001		<0.001	
BVA by WHO								
Normal range	Ref		Ref		Ref		Ref	
Mild impairment	1.49 (1.48-1.51)	<0.001	1.25 (1.24-1.26)	<0.001	1.15 (1.14-1.17)	<0.001	1.12 (1.11-1.14)	<0.001
Moderate impairment	1.94 (1.93-1.95)	<0.001	1.41 (1.40-1.42)	<0.001	1.26 (1.25-1.27)	<0.001	1.20 (1.19-1.21)	<0.001
Severe impairment	1.66 (1.63-1.68)	<0.001	1.28 (1.27-1.30)	<0.001	1.17 (1.15-1.19)	<0.001	1.14 (1.12-1.15)	<0.001
P for trend	<0.001		<0.001		<0.001		<0.001	

^aModel 1: unadjusted. ^bModel 2: adjusted for age and sex. ^cModel 3: adjusted for age, sex, education level, marital status, smoking status, alcohol consumption, BMI, and exercise frequency. ^dModel 4: adjusted for age, sex, education level, marital status, smoking status, alcohol consumption, BMI, exercise frequency, hypertension, diabetes, dyslipidemia, cardiovascular disease, nervous system disease, and city.

Q1 = ≤25th percentile; Q2 = >25th to ≤50th percentile; Q3 = >50th to ≤75th percentile; Q4 = >75th percentile; Ref = reference; other abbreviations as in Table 2.

(RCS) with knots at the 5th, 35th, 65th, and 95th percentiles of the VA distribution. HRs, subdistribution HRs (sHRs), and 95% CIs were presented and forest plots were depicted. Kaplan-Meier cumulative mortality probability curves [1–S(t)] were depicted to

visualize mortality risk across VA categories. Proportional hazards assumption was assessed by log-log survival plots, and no violation was detected for different VA metrics. Sensitivity analyses were conducted by using alternative VA metrics (the presenting

FIGURE 3 Restricted Cubic Spline Curves

Restricted cubic spline curves showing the dose-response relationships between BVA and mortality. (A) BVA and CVD mortality. (B) BVA and cancer mortality. (C) BVA and all-cause mortality. To characterize the shape of the association between continuous VA and mortality risk. Higher BVA values (logMAR) represent worsening visual acuity. A significant nonlinear association was observed for CVD and all-cause mortality, characterized by a steep increase in risk at lower levels of visual impairment which plateaus at higher impairment levels. The trajectory for cancer mortality remained relatively flat. The plateau-shaped curve suggests a potential interplay of risk saturation effect and survivor selection bias in this elderly cohort. Abbreviations as in [Figure 1](#).

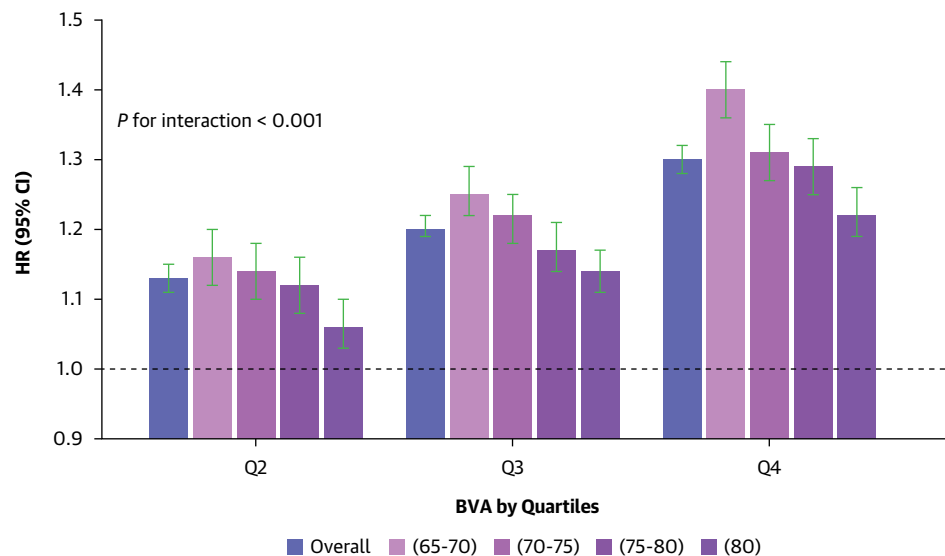
visual acuity of the worse-seeing eye [WVA] and the mean binocular presenting visual acuity [MVA]), repeating the analysis in a multiply imputed dataset, censoring deaths within 12 or 24 months, and recategorizing BMI by WHO criteria (underweight [<18.5 kg/m²], normal weight [≥ 18.5 to <25 kg/m²], overweight [≥ 25 to <30 kg/m²], and obesity [≥ 30 kg/m²]).²⁶ Specifically, multiple imputation was restricted to participants with available VA data ($n = 2,871,348$). To ensure a conservative assessment of robustness, the imputation model used only the covariates from model 4, avoiding the inclusion of exposure (VA) or outcome (mortality) as predictors. To further substantiate the robustness of our findings, we conducted 2 additional sensitivity analyses: excluding the Zaozhuang site due to its substantial participant exclusion rate ($n = 2,285,932$), and performing a

stratified analysis among participants with available ocular disease data ($n = 2,511,951$).

Statistical analyses were performed using SAS version 9.4 (SAS Institute Inc) and R version 4.5.1 (R Foundation for Statistical Computing). All reported P values were 2-tailed. Differences with a $P < 0.05$ were considered statistically significant.

RESULTS

BASELINE CHARACTERISTICS. Baseline characteristics of the study population by BVA categories per WHO criteria and by quartiles are presented in [Table 2](#) and [Supplemental Table 3](#), respectively. In the total of 2,517,662 older participants (mean age: 70.67 ± 6.03 years), 1,071,037 (42.5%) had impaired BVA with 343,926 (13.7%) mild impaired, 634,338 (25.2%)

FIGURE 4 Stratified Analysis of the Association Between BVA and CVD Mortality Across Age Groups

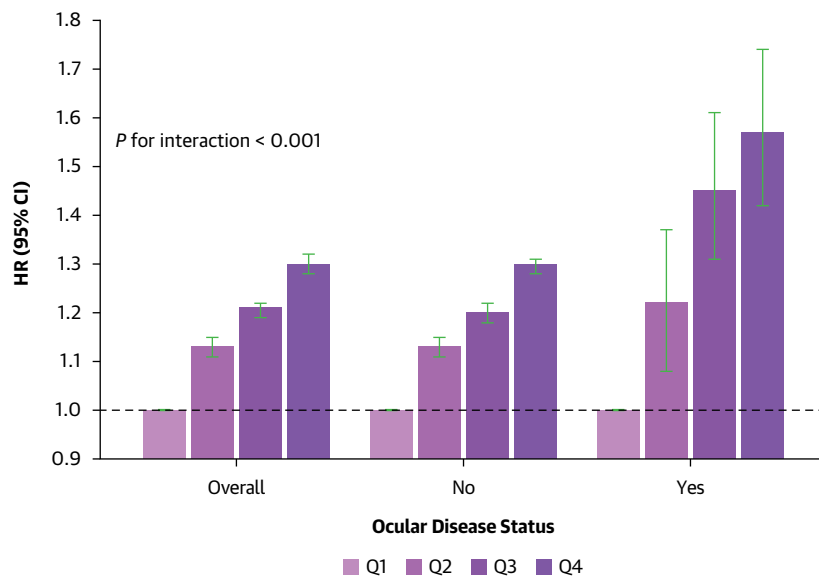
This analysis evaluates whether the association between visual acuity and CVD mortality is modified by age. Higher quartiles (from Q1 to Q4) represent worsening visual acuity. The association was significantly modified by age, being most pronounced in the youngest age group (65-70 years) and attenuating with advancing age. This age-dependent attenuation implies that visual impairment may serve as a more potent and sensitive marker of premature biological aging in younger elderly individuals compared to the oldest-old. Abbreviations as in [Figure 1](#).

moderate impaired, and 92,773 (3.7%) severe impaired. The overall mean BVA was 0.46 ± 0.75 . When stratified by WHO criteria, the mean BVA was 0.16 ± 0.12 in the normal range group, 0.40 ± 0.00 in the mild impairment group, 0.66 ± 0.17 in the moderate impairment group, and 3.94 ± 1.08 in the severe impairment group. When stratified by quartiles, mean BVA was 0.11 ± 0.10 in Q1, 0.30 ± 0.00 in Q2, 0.44 ± 0.05 in Q3, and 1.35 ± 1.34 in Q4. Compared with persons with normal BVA, subjects with impaired BVA were more likely to be older, female, have a lower education level, a higher ratio of single, lower BMI, and less exercise. Over a mean follow-up of 4.99 ± 2.07 years (12,560,718.9 person-years), 348,501 (13.8%) deaths were identified, including 153,758 (44.1%) due to CVD and 68,951 (19.8%) due to cancer. The average mortality rates for all-cause mortality, CVD, and cancer were 27.7, 12.2, and 5.5 per 1,000 person-years, respectively. Elders with impaired BVA had higher mortality rate ([Figure 1](#)).

COX ANALYSIS. Kaplan-Meier cumulative mortality probability curves across BVA categories are presented in [Figure 2](#). The stratification of curves was more apparent with the quartile-based grouping compared to the WHO categorization. Notably, under the WHO criteria, the mortality curve for the

moderate impairment group surpassed that of the severe impairment group. Furthermore, the curves for cancer mortality exhibited a substantial overlap.

Cox analysis results for BVA and mortality are presented in [Table 3](#) and [Supplemental Table 4](#). In the unadjusted model 1, per unit increase in BVA (greater visual impairment) as a continuous variable was associated with a 16% higher risk of CVD mortality (HR: 1.16; 95% CI: 1.16-1.17) and a 14% higher risk of all-cause mortality (HR: 1.14; 95% CI: 1.13-1.14). When BVA was categorized by quartiles, a clear graded association was observed for both CVD and all-cause mortality, with the worst quartile (Q4) showing substantially elevated risks (CVD HR: 2.45; 95% CI: 2.42-2.49; all-cause HR: 2.19; 95% CI: 2.17-2.21; P for trend < 0.001) compared to the best quartile (Q1). When BVA was categorized based on WHO criteria, although trend tests were statistically significant, the moderate impairment group associated with the highest risks (CVD HR: 2.10; 95% CI: 2.07-2.12; all-cause HR: 1.94; 95% CI: 1.93-1.95). In the fully adjusted model 4, which accounted for demographic, lifestyle, clinical, and geographic covariates, the associations were attenuated but remained statistically significant. Per unit increase in continuous BVA (greater visual impairment) was associated with a 3% higher risk of CVD mortality (HR: 1.03; 95% CI: 1.02-1.03) and a 3%

FIGURE 5 Stratified Analysis of the Association Between BVA and CVD Mortality by Ocular Disease Status Among Participants With Available Ocular Disease Information

This analysis evaluates whether the association between VA and CVD mortality is modified by the ocular disease status. Higher quartiles (from Q1 to Q4) represent worsening VA. The association was significantly modified by ocular disease status, being markedly more pronounced in participants with pre-existing ocular disease compared to those without. This interaction suggests that underlying ocular pathology amplifies the adverse association between visual impairment and CVD mortality. Abbreviations as in [Figure 1](#).

higher risk of all-cause mortality (HR: 1.03; 95% CI: 1.03-1.04). Similarly, the worst quartile (Q4) exhibited significantly higher risks of mortality (CVD HR: 1.30; 95% CI: 1.28-1.32; all-cause HR: 1.25; 95% CI: 1.24-1.26) compared to best quartile (Q1). And moderate impairment maintained significantly elevated mortality risks (CVD HR: 1.24; 95% CI: 1.23-1.26; all-cause HR: 1.20; 95% CI: 1.19-1.21). Although all P for trend values remained less than 0.001, the patterns observed in the Kaplan-Meier curves were mirrored in the Cox models. Under the WHO categorization, the severe group exhibited consistently lower HRs than moderate impairment group across all models for CVD and all-cause mortality. In contrast, the associations with cancer mortality were substantially weaker and often failed to reach statistical significance. The analysis of BVA as a continuous variable showed no association with cancer mortality after full adjustment (HR: 1.00; 95% CI: 0.99-1.01; $P = 0.528$). Similarly, using WHO criteria, the severe group showed no significant risk in the unadjusted or fully adjusted models (models 1 and 4), whereas an inverse association was observed in the intermediate models (models 2 and 3).

Dose-response analysis revealed significant nonlinear association between VA and mortality

(P for overall < 0.001; P for nonlinear < 0.001) ([Figure 3](#)). RCS curves indicated that both CVD mortality and all-cause mortality risks rose steeply with initial deterioration in VA before the risk elevation attenuated and eventually plateaued at more severe impairment levels. In contrast, the trajectory for cancer mortality remained relatively flat.

STRATIFIED ANALYSIS. Stratified analyses showed the robustness of the association between BVA and CVD mortality across diverse subgroups ([Supplemental Table 5](#)). After full adjustment, significantly elevated risks were consistently observed in all strata ($P < 0.001$). Significant effect modification was identified for age, sex, and city (P for interaction < 0.001). Specifically, the association was most pronounced in the youngest age group (65-70 years, HR for Q4: 1.40; 95% CI: 1.36-1.44) and attenuated with advancing age ([Figure 4](#)). Similarly, the association appeared slightly stronger in females than in males. Geographic heterogeneity was also evident; participants in Luzhou (HR: 1.49; 95% CI: 1.45-1.52) and Shenzhen (HR: 1.43; 95% CI: 1.36-1.50) exhibited relatively larger risk estimates compared to those in Zunyi (HR: 1.22; 95% CI: 1.19-1.24) and Zaozhuang (HR: 1.16; 95% CI: 1.12-1.22). In contrast,

TABLE 4 Competing Risk Analysis of the Association Between Different VA Metrics and Cause-Specific Mortality (CVD and Cancer)^a

Category	BVA		WVA		MVA	
	HR (95% CI)	P Value	HR (95% CI)	P Value	HR (95% CI)	P Value
CVD mortality						
VA as continuous	1.03 (1.02-1.03)	<0.001	1.03 (1.02-1.03)	<0.001	1.03 (1.02-1.03)	<0.001
VA by quartiles						
Q1	Ref		Ref		Ref	
Q2	1.13 (1.11-1.15)	<0.001	1.13 (1.11-1.15)	<0.001	1.12 (1.10-1.13)	<0.001
Q3	1.19 (1.18-1.21)	<0.001	1.21 (1.19-1.23)	<0.001	1.19 (1.17-1.21)	<0.001
Q4	1.28 (1.27-1.30)	<0.001	1.28 (1.27-1.30)	<0.001	1.28 (1.26-1.30)	<0.001
VA by WHO						
Normal range	Ref		Ref		Ref	
Mild impairment	1.12 (1.10-1.14)	<0.001	1.11 (1.09-1.13)	<0.001	1.12 (1.10-1.13)	<0.001
Moderate impairment	1.22 (1.21-1.24)	<0.001	1.21 (1.20-1.23)	<0.001	1.23 (1.22-1.24)	<0.001
Severe impairment	1.16 (1.14-1.18)	<0.001	1.18 (1.16-1.20)	<0.001	1.18 (1.16-1.21)	<0.001
Cancer mortality						
VA as continuous	1.00 (0.99-1.01)	0.550	1.00 (0.99-1.01)	0.540	1.00 (0.99-1.01)	0.530
VA by quartiles						
Q1	Ref		Ref		Ref	
Q2	1.05 (1.03-1.07)	<0.001	1.05 (1.03-1.07)	<0.001	1.05 (1.03-1.07)	<0.001
Q3	1.06 (1.04-1.08)	<0.001	1.07 (1.04-1.09)	<0.001	1.07 (1.05-1.10)	<0.001
Q4	1.02 (1.00-1.05)	0.041	1.03 (1.01-1.06)	0.007	1.04 (1.02-1.06)	<0.001
VA by WHO						
Normal range	Ref		Ref		Ref	
Mild impairment	1.06 (1.03-1.08)	<0.001	1.05 (1.03-1.07)	<0.001	1.06 (1.04-1.08)	<0.001
Moderate impairment	1.02 (1.00-1.03)	0.120	1.04 (1.02-1.06)	<0.001	1.03 (1.01-1.05)	0.003
Severe impairment	1.00 (0.96-1.05)	0.940	1.01 (0.97-1.05)	0.570	1.01 (0.97-1.05)	0.560

^aAdjusted for age, sex, education level, marital status, smoking status, alcohol consumption, BMI, exercise frequency, hypertension, diabetes, dyslipidemia, cardiovascular disease, nervous system disease, and city.
VA = visual acuity; other abbreviations as in [Tables 2 and 3](#).

no significant interaction was observed for baseline hypertension ($P_{\text{for interaction}} = 0.670$), diabetes ($P_{\text{for interaction}} = 0.395$), dyslipidemia ($P_{\text{for interaction}} = 0.109$), or CVD ($P_{\text{for interaction}} = 0.481$).

Regarding ocular disease status, the association remained virtually unchanged in those without ocular disease but was stronger in participants with preexisting ocular disease (HR for Q4: 1.57; 95% CI: 1.42-1.74; $P_{\text{for interaction}} < 0.001$) ([Figure 5](#), [Supplemental Table 6](#)).

COMPETING RISK ANALYSIS. Competing risk analyses produced results highly consistent with the primary Cox models ([Table 4](#)). Specifically, per unit increase in continuous BVA (greater visual impairment) was associated with a 3% higher CVD mortality risk (sHR: 1.03; 95% CI: 1.02-1.03), and Q4 exhibited a 28% elevated risk vs Q1 (sHR: 1.28; 95% CI: 1.27-1.30). In contrast, associations with cancer mortality remained substantially weaker and failed to reach statistical significance.

SENSITIVITY ANALYSIS. Extensive sensitivity analyses further corroborate the primary results. When missing covariates data were handled using multiple

imputation, the HRs were modestly higher than those in the primary analysis, with an HR of 1.04 (95% CI: 1.03-1.04) for CVD mortality and 1.04 (95% CI: 1.03-1.04) for all-cause mortality per unit increase in BVA (greater visual impairment) ([Supplemental Table 7](#)). Analyses that censored early deaths (within the first 12 or 24 months) or recategorized BMI by WHO criteria yielded results nearly identical to primary analyses, with a fully adjusted HR for CVD mortality of 1.03 (95% CI: 1.02-1.03) per unit increase in BVA (greater visual impairment) in all scenarios ([Supplemental Tables 8 to 10](#)). Additionally, dose-response trends remained statistically significant (all $P_{\text{for trend}} < 0.001$). Furthermore, our findings remained robust regardless of the specific VA metric used. Kaplan-Meier cumulative mortality curves and RCS curves for WVA and MVA are shown in [Supplemental Figures 2 to 5](#). After full adjustment, the magnitude of risk estimates for BVA, WVA, and MVA was virtually identical across all analytical scenarios ([Table 4](#), [Supplemental Tables 6 to 14](#)). Results remained unchanged after excluding the Zaozhuang site ([Supplemental Table 15](#)).

DISCUSSION

In this multicenter, city-level, large-scale cohort study, we found that poor VA was independently associated with elevated risks of both CVD and all-cause mortality in a graded trend. This association was robust across extensive subgroup analyses and a series of sensitivity analyses, including the use of different VA metrics, censoring early deaths, repeating analyses in a multiply imputed data set, and using competing-risk models to account for cause-specific mortality.

CONTEXTUALIZATION WITH PREVIOUS EVIDENCE AND POTENTIAL INSIGHTS. The observed association between VA and CVD mortality is biologically and functionally plausible, supported by concepts from the emerging field of oculosimics⁷ and the conceptual framework of the Lancet Global Health Commission.⁴ Biologically, the eye serves as a unique, noninvasive window into the body's systemic cardiovascular health. As highlighted in recent oculosimics research, the retina's extensive microvascular network shares embryological origins and physiological properties with coronary and cerebral circulations.⁵⁻⁷ Moreover, as shown by the Lancet Commission's review, eye health and cardiovascular health are inextricably linked through shared traditional risk factors, such as hypertension, diabetes, and smoking.⁸⁻¹¹ Furthermore, establishing a bidirectional pathological link, CVD such as hypertension, stroke, arteriosclerosis, and microcirculatory dysfunction can directly compromise visual function through vascular pathology.²⁷ Beyond these pathophysiological mechanisms, VA impairment acts as a functional accelerator of health deterioration.^{4,28} Poor vision precipitates a cascade of adverse outcomes—including restricted physical activity and social engagement, injuries, depression, cognitive decline, and reduced access to health care—which collectively exacerbate physical frailty and shorten life expectancy. Crucially, the observation that associations with cancer mortality remained substantially weaker and failed to reach statistical significance distinguishes VA from a generic marker of frailty, underscoring its predominant role as a specific sentinel for cardiovascular health rather than generalized sickness. Therefore, VA functions as an integrative indicator of systemic vascular health, reflecting biological aging by capturing both the history of vascular injury and the depletion of functional reserve.

Stratified analyses provide empirical support for the mechanistic framework described above.

The association between VA and CVD mortality persisted regardless of baseline clinical history (including hypertension, diabetes, dyslipidemia, or prior CVD), suggesting that VA impairment contributed to cardiovascular risk through pathways distinct from, or additive to, traditional cardiometabolic mechanisms. Beyond individual clinical profiles, geographic heterogeneity also played a modulatory role. The observation of relatively stronger associations in Luzhou and Shenzhen compared to Zunyi and Zaozhuang implies that regional determinants—such as varying health care accessibility, socioeconomic contexts, or environmental exposures—may interact with VA impairment to amplify or attenuate its prognostic impact.

Notably, the prognostic value of VA was significantly modified by age. The association was most pronounced in the youngest subgroup (65-70 years) and attenuated with advancing age. This trend is corroborated by the Kangbuk Samsung Health Study,¹⁵ which reported a much stronger association in a substantially younger and generally healthier population (average age: 39.7 years). The effect size in that younger cohort substantially exceeded that in our elderly population—a 124% vs 3% increase in CVD mortality risk per 1-logMAR unit increase (greater visual impairment). This striking disparity underscores the concept of risk saturation in older age: VA impairment serves as a potent, sensitive marker of premature biological aging in younger populations; conversely, in older adults, its prognostic specificity is partially attenuated by its high prevalence (42.5% in our cohort vs 6.6% in the Korean cohort) and integration within a broader multimorbidity profile.

Consistent with this age-dependent attenuation, the dose-response relationship between VA impairment and CVD mortality in our cohort also manifested as a plateau-shaped nonlinear curve. This plateau-shaped pattern provides critical epidemiological insights, likely reflecting the interplay of risk saturation and survivor selection in this elderly cohort. First, the plateau exemplifies the phenomenon of risk saturation.^{29,30} As VA impairment advances, the marginal contribution of further vision loss to mortality diminishes because the association is overshadowed by age-related comorbidities (eg, frailty, cognitive decline, or end-stage chronic diseases), which may become the predominant mortality drivers. Additionally, a pathophysiological threshold may exist beyond which additional visual damage no longer proportionally indicates the systemic vascular degeneration underlying the vision-

mortality association. Second, this nonlinearity is consistent with survivor selection bias.^{31,32} Individuals who have survived to advanced age despite severe VA impairment may represent a biologically robust subgroup with greater physiological reserve or adaptive capacity, partially buffering the lethal impact of their ocular condition, thereby flattening the risk curve at the severe end of the spectrum. Finally, from a measurement perspective, the precision of VA assessment may be reduced at the severe end, potentially attenuating the observed risk gradient through misclassification.¹²

Previous meta-analyses and studies have suggested that the association between VA and mortality may vary by the underlying cause of impairment.^{12,13,16,33,34} Specifically, nonrefractive causes (eg, cataract, retinopathy, and macular degeneration, reflecting underlying pathology) might have a stronger association than correctable refractive errors. For instance, pooled analyses have shown attenuated effect sizes in studies using presenting VA compared to those using best-corrected VA.¹² Similarly, evidence from the China Health and Retirement Longitudinal Study indicated that although VA impairment increases mortality, visual correction mitigates this risk.³⁴ In the current study, VA was assessed as presenting VA within the framework of Basic Public Health Services project. Although presenting VA offers advantages for large-scale surveys by capturing real-world functional capacity and identifying unmet eye care needs, the absence of data on refractive correction status during assessment precludes a direct distinction between correctable refractive errors and uncorrectable impairment. To mitigate the impact of this information gap and better delineate the role of underlying pathology, we stratified our analysis by ocular disease status. A significant interaction was observed: the association between VA impairment and CVD mortality was markedly more pronounced in participants with preexisting ocular disease compared to those without. This finding aligns with the literature, suggesting that underlying ocular pathology amplifies the adverse association between VA impairment and CVD mortality, potentially by reflecting a heavier burden of systemic vascular damage.

Regarding different VA metrics (better eye, worse eye, or binocular), our findings have indicated that the magnitude of risk estimates for BVA, WVA, and MVA was virtually identical across all analytical scenarios after full adjustment. This consistency aligns with the previous meta-analysis, where meta-regression showed no significant variation in effect size by the eye assessed ($P = 0.24$). This equivalence

is clinically and epidemiologically reassuring. In public health assessments, BVA remains the most intuitive metric as it aligns with WHO standards and directly reflects an individual's functional capacity for daily living. However, given that the acuity of both eyes is routinely recorded during standard VA assessments, data on WVA or MVA are often readily available and may offer complementary insights into pathological burdens or integrated visual function. Ultimately, the comparable risk estimates across metrics reinforce the robustness of VA as a generalized functional signal of systemic vascular health, highlighting a prognostic value intrinsic to visual function rather than contingent upon a specific measurement choice.

Quartile-based VA categorization has shown superior risk stratification compared to the WHO criteria in both Kaplan-Meier and Cox analyses within the current study, suggesting that an individual's relative VA ranking within their age cohort may serve as a more sensitive survival predictor than absolute, universal cut-offs.

PUBLIC HEALTH AND CLINICAL IMPLICATIONS. Our findings, which identify VA impairment as a significant and independent risk marker for CVD mortality, carry important implications for both public health strategy and clinical practice. Although the effect size for continuous VA appeared modest, the categorical presence of VA impairment was associated with a substantially elevated risk of CVD mortality. Given the high prevalence of VA impairment in the elderly, this represents a considerable population-level burden, even after adjustment for traditional risk factors.

The strategic significance of these findings extends beyond individual well-being to align with fundamental goals of global and national public health agendas. The persistent association suggests VA not merely as an indicator of ocular health but as a potential barometer of systemic cardiovascular integrity and biological aging. This resonates with the WHO's definition of healthy aging as "the process of developing and maintaining the functional ability that enables wellbeing in older age."³⁵ By framing VA as a critical component of functional ability, our evidence directly supports the "Decade of Healthy Aging 2021-2030".³⁶ and China's "Healthy China 2030" initiative,³⁷ which explicitly emphasize strengthening comprehensive health management for the elderly.³⁸

In the context of rapid population aging, our findings suggest that VA should be interpreted as a readily observable functional marker of systemic vulnerability rather than a determinant for specific

cardiovascular intervention. Impaired VA signals broader deficits in physiological reserves and the capacity to manage health, prompting physicians to contextualize cardiovascular risk within the wider framework of functional aging rather than relying solely on traditional risk factors. From a practical standpoint, poor vision acts as a barrier to health system navigation and treatment adherence. Consequently, patients with VA impairment warrant tailored accommodation such as simplified communication, enhanced caregiver support, or fall prevention strategies to overcome these functional hurdles. Integrating such supportive measures into routine practice represents a feasible step towards the goals of both the Decade of Healthy Aging 2020-2030 and Healthy China 2030 initiative, shifting the focus from disease-specific treatment to holistic functional preservation and contributing to the extension of healthy life expectancy.³⁹

To our knowledge, this represents the first and largest investigation specifically examining the association between VA and CVD mortality among older adults in China, filling a critical evidence gap in this rapidly aging population. Our mega-cohort of >2.52 million participants from multiple cities provides exceptional statistical power and enhances the generalizability of our findings. Methodologically, our approach was comprehensive and rigorous. We used multiple adjustment models (from crude to fully adjusted with demographic, lifestyle, geographic factors, and baseline clinical condition) to systematically control for potential confounders. The exposure was analyzed through different stratification strategies (WHO criteria and population-based quartiles) to capture the association pattern thoroughly. We also conducted extensive subgroup analyses (eg, by age, sex, and baseline clinical conditions) and sensitivity analyses, including the use of different VA metrics (better-eye, worse-eye, and mean binocular acuity), repeating analyses in a multiply imputed dataset, and using competing-risk models to account for cause-specific mortality. To address the potential issue of reverse causality, we conducted additional sensitivity analyses censoring early deaths (within 12 or 24 months) and results were consistent. The application of RCS curves further confirmed a nonlinear dose-response relationship. Moreover, all data were derived from a standardized national public health service project and an authoritative national mortality surveillance system, ensuring the reliability of both VA measurements and mortality ascertainment.

STUDY LIMITATIONS. First, a major limitation is the lack of recorded information on corrective lens status or received ocular treatment. This data deficit precluded the distinction between correctable refractive errors and uncorrectable VA impairment and prevented us from analyzing the potential impact of corrective or therapeutic interventions on the observed association. Although we addressed this to some extent by stratifying participants based on general ocular disease status, the lack of granular diagnostic data (eg, cataract, diabetic retinopathy, or age-related macular degeneration) limited our ability to explore how specific ocular pathologies might differentially modify the association with CVD mortality. Future studies incorporating specific ocular diagnoses and comprehensive intervention-related data are therefore warranted to better identify which ocular pathologies are most closely linked to cardiovascular mortality and systemic vascular burden and to evaluate the potential impact of therapeutic interventions. Second, our assessment was limited to VA. We could not evaluate other multidimensional aspects of visual function such as visual field or contrast sensitivity, which are also important for daily functioning and may be independently associated with mortality. Future investigations could supplement these findings with multidimensional assessments of visual function to capture a more holistic profile of visual health. Third, our study was limited to mortality outcomes and lacked data on incident CVD events, which prevented us from elucidating the specific underlying pathway—whether VA impairment contributed to CVD mortality by increasing disease incidence, exacerbating post-onset prognosis, or both. To address this, future longitudinal studies incorporating the monitoring of incident events are recommended to disentangle these mechanisms. Furthermore, supplementing such research with analyses on the potential impact of unincluded variables would significantly strengthen the mechanistic understanding of the link between visual function and cardiovascular outcomes. Fourth, despite extensive adjustments, the potential for residual confounding due to unmeasured or imperfectly measured factors cannot be eliminated. Notably, lifestyle factors (eg, smoking status, alcohol consumption, and exercise frequency) were ascertained via self-reports; given their inherent susceptibility to recall bias or social desirability bias, residual confounding arising from measurement error remains a potential concern. Fifth, although the Basic Public Health Services project

covers both urban and rural residents, the specific urban-rural boundaries within the dataset were often ambiguous, precluding a distinct stratification at this level. However, we mitigated this by conducting city-stratified analyses. Given the marked socioeconomic heterogeneity and varying medical accessibility among the four cities—reflecting a spectrum from the predominantly urbanized population of Shenzhen to the substantial rural demographics of Luzhou and Zunyi—these analyses have successfully accounted for substantial socioeconomic variations. Sixth, site-specific variations in data completeness, particularly the high exclusion rate in Zaozhuang, raise the possibility of selection bias. However, the consistent results from sensitivity analyses, including multiple imputations and excluding the Zaozhuang site, affirm the robustness of our findings against missing data. Finally, given the notable variations in effect sizes observed across our study sites and the substantial socioeconomic diversity inherent to China's extensive territory, caution is still warranted when generalizing these findings. Extrapolation to regions with significantly different demographic or economic profiles should be approached with care.

CONCLUSIONS

In this large-scale cohort study of Chinese older adults, we found that VA impairment was highly prevalent and demonstrated a significant, independent, and graded association with elevated risks of both all-cause and CVD mortality. Crucially, our findings reframe VA beyond its traditional ocular role, positioning it as a readily observable functional signal of cardiovascular vulnerability. Rather than implying a direct target for cardiovascular intervention, this signal warrants a shift in clinical and public health paradigms: from viewing vision loss in isolation to contextualizing it within the broader framework of physiological reserve, systemic vulnerability, and biological aging. Such a perspective prompts

physicians to situate cardiovascular risk within the broader landscape of functional aging, moving beyond reliance solely on traditional risk factors. Ultimately, prioritizing such holistic functional preservation contributes to the strategic goals of the “Healthy China 2030” initiative and the extension of healthy life expectancy in China and globally.

DATA AVAILABILITY Data included in this study are not publicly available. The National Center for Chronic and Noncommunicable Disease Control and Prevention, Chinese Center for Disease Control and Prevention provides conditional data access for qualified researchers with legitimate requests. Please contact corresponding author to seek approval for data access.

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APPENDIX For additional tables and figures, please see the online version of this paper.
